

Image Quality Check SOP

For AI Defect Detection and Root-Cause Assistant

MVTec AD Category Scope: bottle, cable, screw, transistor, tile, toothbrush, zipper, metal_nut

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Important note: This document is a simulated industrial quality-control SOP created for a student prototype and RAG knowledge base. It is not an official manufacturer document and must not be used as a real factory safety or compliance standard.

1. Document Control

This Image Quality Check SOP defines the minimum image requirements before an uploaded product image is accepted by the AI inspection pipeline. The goal is to prevent poor images from creating false defect predictions, weak RAG explanations, and unreliable inspection reports.

Field	Value
Owner	Quality AI System Team
Applies To	Images uploaded for MVTec AD categories: bottle, cable, screw, transistor, tile, toothbrush, zipper, metal_nut
Users	Line operator, QA inspector, supervisor, AI system administrator
System Modules Affected	Frontend upload, image preprocessing, computer vision inference, OCR, RAG, LLM explanation, PDF report generation
Review Frequency	Update when product categories, camera setup, model thresholds, or defect classes change

2. Purpose

The purpose of this SOP is to make sure every input image is clear, complete, and useful before the AI model performs defect detection or anomaly analysis. Good image quality improves model accuracy, reduces false positives and false negatives, and helps the RAG system produce more reliable explanations.

The system must not treat every uploaded image as valid. If the image is blurry, too dark, overexposed, cropped, strongly reflected, compressed, rotated incorrectly, or does not show the selected product, the system should request a retake or send the case for human review.

3. Scope

This SOP applies to image uploads and video-frame captures used in the AI Defect Detection and Root-Cause Assistant. It covers the eight selected MVTec AD categories used in this project: bottle, cable, screw, transistor, tile, toothbrush, zipper, and metal_nut.

- Applies to single product images uploaded through the web interface.
- Applies to frames extracted from recorded or live video streams in future versions.
- Applies before computer vision model prediction, OCR, RAG retrieval, and report generation.
- Does not define final defect severity. It only decides whether the image is suitable for inspection.

4. Key Definitions

Term	Meaning in this project
Image Quality Check	A validation step that checks whether an image is suitable for AI inspection before prediction.
PASS	The image is clear enough for model inference and can move to defect detection.
RETAKE_IMAGE	The image is not suitable. The user must upload or capture a better image.
HUMAN_REVIEW	The image has borderline quality or unusual conditions. A human inspector should verify.
Region of Interest (ROI)	The product area that must be visible for inspection.
Blur	Loss of sharpness caused by poor focus, motion, or camera shake.
Exposure	Brightness level of the image. Underexposure is too dark; overexposure is too bright.
Occlusion	The product is partly hidden by another object, hand, label, shadow, or cropping.
Reflection/Glare	Bright spots that hide the real product surface.

Term	Meaning in this project
Category Mismatch	The uploaded product does not match the selected category. Example: user selects bottle but uploads tile.

5. Image Quality Decision Codes

Code	Decision	Meaning	Next Action
PASS	Accept image	Image is suitable for AI inspection.	Run preprocessing and computer vision inference.
PASS_WITH_WARNING	Accept with caution	Small quality issue exists but inspection may still be useful.	Run inference and mark report with image quality warning.
RETAKE_IMAGE	Reject image	Image quality is too poor for reliable inspection.	Ask operator to retake the image with corrected guidance.
HUMAN_REVIEW	Escalate	Image is borderline, unusual, or system cannot confidently validate it.	Send to QA inspector or supervisor before final decision.

6. Standard Image Quality Requirements

The following quality dimensions should be checked automatically by the system and manually by the operator when needed.

Quality Factor	Accepted Condition	Reject / Retake Condition	Why it matters
Focus / Sharpness	Product edges and important surface details are visible.	Image is blurry, shaken, or product boundary is unclear.	Blur can hide cracks, small scratches, thread damage, cable cuts, and transistor pin defects.
Brightness	Product is visible with balanced lighting.	Image is too dark or too bright.	Bad brightness can make normal surfaces look defective or hide real defects.
Contrast	Defect regions can be separated from background.	Low contrast makes product and background look similar.	Poor contrast weakens segmentation and anomaly localization.
Framing	Full product or target ROI is visible.	Product is cropped, partly outside the image, or too small.	A cropped product prevents complete inspection.
Occlusion	No hand, tool, shadow, label, or fixture hides the product.	Important inspection area is blocked.	Occlusion may be incorrectly detected as a defect.
Reflection / Glare	Surface details remain visible.	Strong glare hides surface or edges.	Glare can look like a crack, stain, or missing region.
Resolution	Enough pixels to inspect surface and structural details.	Very small, pixelated, or heavily compressed image.	Low resolution reduces confidence and may create false decisions.
Orientation	Product orientation is consistent with training or expected view.	Rotated, flipped, or unusual view without clear product context.	Unusual orientation can reduce model performance.
Background	Simple, stable background with limited distractions.	Cluttered background or multiple objects confuse ROI detection.	Background noise can trigger false anomaly regions.
File Integrity	Readable image file in supported format.	Corrupted file, unsupported type, or failed decode.	System cannot inspect invalid files.

7. Recommended Automated Validation Checks

The following checks can be implemented in Python before calling the vision model. Thresholds should be tuned after testing your selected dataset and camera conditions. These values are starter guidance for a prototype, not fixed factory standards.

Check	Example Method	Starter Threshold / Rule	Decision Logic
File type	Extension and MIME validation	Allow JPG, JPEG, PNG, BMP, WEBP	Invalid type -> RETAKE_IMAGE
File size	Frontend/backend size check	Recommended 50 KB to 10 MB	Too small or too large -> warning or reject
Image resolution	width x height	Minimum 224 x 224 for classifier; prefer 512 x 512 or higher for localization	Below minimum -> RETAKE_IMAGE
Blur score	Variance of Laplacian	If score is very low, image is likely blurry	Low sharpness -> RETAKE_IMAGE
Brightness mean	Mean grayscale intensity	Very low = too dark; very high = too bright	Extreme brightness -> RETAKE_IMAGE
Contrast	Standard deviation of grayscale intensity	Very low contrast indicates flat/unusable image	Low contrast -> PASS_WITH_WARNING or RETAKE_IMAGE
Saturation / glare	Bright pixel percentage and white clipping	Large overexposed region near product indicates glare	Strong glare -> RETAKE_IMAGE
Product occupancy	ROI area / image area	Product should occupy a useful portion of image	Too small or cropped -> RETAKE_IMAGE
Duplicate image	Hash comparison	Detect repeated uploads	Duplicate can be accepted but marked in report
Category plausibility	Lightweight classifier or user selection check	Uploaded product should match selected category	Mismatch -> HUMAN_REVIEW or ask user to correct category

7.1 Example validation decision logic

The implementation can combine multiple quality checks instead of relying on one score.

```
quality_status = PASS
if file_type_invalid or image_decode_failed:
    quality_status = RETAKE_IMAGE
elif resolution_below_minimum:
    quality_status = RETAKE_IMAGE
elif blur_score_too_low:
    quality_status = RETAKE_IMAGE
elif extreme_darkness or extreme_overexposure:
    quality_status = RETAKE_IMAGE
elif product_cropped or product_not_visible:
    quality_status = RETAKE_IMAGE
elif low_contrast or moderate_glare:
    quality_status = PASS_WITH_WARNING
elif category_mismatch_uncertain:
    quality_status = HUMAN_REVIEW
else:
    quality_status = PASS
```

8. Operator Capture SOP

The operator must follow the capture steps below when taking a new image or retaking a rejected image.

Step	Operator Instruction	Reason
1	Place the product on a clean, stable background.	Reduces background noise and false detections.
2	Use stable lighting from more than one direction where possible.	Reduces shadows and overexposed glare.
3	Keep the camera steady and focus on the product surface.	Prevents blur and unclear edges.

Step	Operator Instruction	Reason
4	Capture the complete product or the full required inspection region.	Prevents cropped defects and missing areas.
5	Avoid touching or covering the product during capture.	Prevents occlusion and false anomaly regions.
6	For reflective surfaces, slightly adjust angle to reduce glare.	Improves visibility of cracks, scratches, dents, and contamination.
7	Upload the image using the correct product category.	Prevents category mismatch and wrong model selection.
8	If the system asks for retake, follow the reason shown by the system.	Improves second attempt quality and reduces repeated failure.

9. Category-Specific Image Quality SOP

Different product categories need different capture and quality checks. The following rules are designed for the selected MVTec AD categories and should be used as RAG guidance and frontend retake messages.

9.1 Bottle Image Quality Requirements

Inspection focus: Inspect shape, neck, rim, cap/contact region, sidewall, bottom edge, contamination marks, cracks, broken surfaces, and foreign particles.

- The full bottle body and rim area must be visible.
- Avoid strong glare on transparent or shiny surfaces.
- Use a plain background so the bottle contour is clear.
- If the bottle is transparent, use backlighting or side lighting to show cracks and contamination.
- Do not crop the neck, rim, or bottom edge when these areas are the inspection target.

Retake rule: RETAKE_IMAGE if glare hides more than a small region of the bottle surface, if the bottle is not fully visible, or if the defect region is outside the frame.

9.2 Cable Image Quality Requirements

Inspection focus: Inspect insulation surface, bent sections, cut marks, exposed wire, contamination, missing cable segments, and connector condition.

- Capture the entire cable segment or connector region being inspected.
- Cable should not overlap itself excessively unless overlap is part of the normal category.
- Use contrast between cable and background so cable boundary is clear.
- Avoid shadows that look like cuts or missing insulation.
- For connectors, include the connector pins and cable entry point.

Retake rule: RETAKE_IMAGE if cable boundary cannot be separated from background, if the connector is cropped, or if dark shadows look like cable damage.

9.3 Screw Image Quality Requirements

Inspection focus: Inspect head, shaft, thread continuity, tip, broken areas, bent shape, contamination, and surface damage.

- Use high sharpness because screw defects are often small.
- Capture the full screw including head, threads, and tip.
- Use close-up resolution but avoid cropping the item.
- Avoid motion blur; use a fixed camera or tripod if possible.

- Use side lighting to reveal damaged threads without overexposing the metallic surface.

Retake rule: RETAKE_IMAGE if threads are blurred, screw tip is missing from image, or metallic reflection hides thread details.

9.4 Transistor Image Quality Requirements

Inspection focus: Inspect body, pins/leads, alignment, bent/missing leads, damaged case, contamination, and marking/label if visible.

- Capture the complete transistor including all pins/leads.
- Keep the component flat and clearly separated from background.
- Avoid glare on black or shiny casing.
- Ensure pin area is in focus because pin damage may be small.
- If OCR is needed, make printed markings readable.

Retake rule: RETAKE_IMAGE if leads are cropped, pin region is out of focus, or casing text needed for OCR is unreadable.

9.5 Tile Image Quality Requirements

Inspection focus: Inspect surface texture, cracks, chips, stains, edge damage, scratches, and pattern irregularities.

- Capture a flat top-down image when possible.
- Show the full tile or clearly define the inspected region.
- Use even lighting to avoid shadows that look like cracks.
- Avoid reflection if the tile surface is glossy.
- For edge damage, include all relevant tile edges.

Retake rule: RETAKE_IMAGE if uneven shadows hide cracks, if tile edges are cropped for an edge-defect inspection, or if surface glare hides texture.

9.6 Toothbrush Image Quality Requirements

Inspection focus: Inspect bristles, brush head, handle, missing bristle groups, deformation, contamination, and broken structure.

- Capture the brush head clearly and keep bristles in focus.
- Use background contrast so bristle tips are visible.
- Capture the complete handle if handle defects are inspected.
- Avoid hand occlusion near bristles.
- Use side angle if bristle deformation is the target defect.

Retake rule: RETAKE_IMAGE if bristles are blurred, hidden, strongly shadowed, or the brush head is cropped.

9.7 Zipper Image Quality Requirements

Inspection focus: Inspect teeth, slider, tape, missing teeth, misalignment, broken parts, contamination, and stitching area.

- Capture the zipper teeth line clearly along its full inspected length.
- Use enough resolution so individual teeth are visible.
- Avoid folds or bends unless they are part of the inspection condition.
- Keep slider and start/end points visible if they are relevant.
- Use even light to show gaps between teeth.

Retake rule: RETAKE_IMAGE if teeth cannot be counted/seen, if the zipper is heavily folded, or if the slider/teeth target region is cropped.

9.8 Metal Nut Image Quality Requirements

Inspection focus: Inspect hole shape, thread condition, edges, cracks, dents, corrosion, machining marks, and surface contamination.

- Capture the full nut, including inner hole and outer edge.
- Use close-up focus for inner threads and edge defects.
- Reduce metallic glare by adjusting camera angle.
- Use plain background for clear contour detection.
- For thread inspection, align the camera so the inner area is visible.

Retake rule: RETAKE_IMAGE if inner hole is dark, threads are blurred, glare hides edges, or the nut is partly outside the image.

10. Image Quality Status Messages for Frontend

The frontend should provide clear retake messages so the user understands exactly what is wrong. Avoid vague messages such as "bad image". Use action-based messages.

Detected Issue	User-Friendly Message	Recommended UI Action
Blur	The image is too blurry. Please retake the photo with the camera focused on the product.	Show retake button and blur reason.
Too dark	The image is too dark. Please improve lighting and upload again.	Show brightness guidance.
Too bright / glare	Strong glare or overexposure is hiding product details. Please change the angle or lighting.	Show glare warning.
Cropped product	The product is not fully visible. Please capture the complete product or target inspection region.	Show framing overlay example.
Low resolution	The image resolution is too low for reliable inspection. Please upload a clearer image.	Block inference unless override allowed.
Wrong category	The uploaded image may not match the selected product category. Please confirm or choose the correct category.	Ask user to confirm category.
Occlusion	Part of the product is hidden. Please remove any hand, tool, shadow, or object covering the product.	Request retake.
Unreadable label	Text on the product is not readable. Please retake if batch or label verification is required.	Allow prediction but OCR marked failed.

11. Integration with AI Pipeline

The image quality module should run before defect detection. Its output must be passed to the agent, RAG system, LLM explanation layer, and report generator.

Pipeline Stage	How Image Quality Output Is Used
Frontend Upload	Shows immediate image quality feedback when possible.
Backend Validation	Rejects invalid files and computes quality metrics.
Preprocessing	Applies resizing, normalization, orientation correction, and optional ROI extraction.
Vision Model	Runs only when quality status is PASS or PASS_WITH_WARNING.
OCR	Runs only if text is visible and image quality is suitable for reading.
Agent Routing	If RETAKE_IMAGE, agent should not call RAG for final defect recommendation. It should ask for a clearer image.

Pipeline Stage	How Image Quality Output Is Used
RAG Retrieval	Retrieves this SOP when user asks why the image was rejected or how to retake it.
LLM Explanation	Explains the image quality issue in simple language.
Report Generation	Stores image quality metrics and warnings in the inspection report.

12. Agent Rules for Image Quality

The AI agent should follow these rules to avoid generating unsupported or unsafe inspection results from poor images.

Condition	Agent Decision
quality_status = PASS	Proceed to vision inference, then call RAG if needed.
quality_status = PASS_WITH_WARNING	Proceed to vision inference but include warning in final answer and report.
quality_status = RETAKE_IMAGE	Do not give final defect decision. Ask user to upload a clearer image and provide specific retake instructions.
quality_status = HUMAN_REVIEW	Do not make an automatic final acceptance/rejection decision. Send to human inspector or supervisor.
OCR required but text unreadable	Run visual defect detection if image is otherwise usable, but mark OCR as failed and ask for closer text image if needed.
Critical-looking defect but poor image	Escalate to human review rather than ignoring the defect or making a confident final decision.

13. Image Quality Metrics for Reports

Every generated inspection report should include an image quality section. This improves traceability and helps explain why the AI accepted, rejected, or escalated an image.

Report Field	Description	Example
image_quality_status	Final quality decision code.	PASS_WITH_WARNING
resolution	Uploaded image width and height.	1024 x 768
blur_score	Computed sharpness score.	185.4
brightness_level	Summary of exposure.	Normal
contrast_level	Summary of contrast.	Acceptable
glare_detected	Whether strong reflection was detected.	No
occlusion_detected	Whether product area is hidden.	No
category_confirmed	Whether image matches selected category.	Yes
quality_warning	Human-readable warning if any.	Minor glare detected but product surface remains visible.
retake_instruction	Specific instruction when rejected.	Retake image with better focus and full product visible.

13.1 Suggested JSON output schema

```
{
  "image_quality": {
    "status": "PASS | PASS_WITH_WARNING | RETAKE_IMAGE | HUMAN_REVIEW",
    "resolution": {"width": 1024, "height": 768},
    "blur_score": 185.4,
    "brightness_status": "normal",
    "contrast_status": "acceptable",
    "glare_detected": false,
```

```

    "occlusion_detected": false,
    "category_match": "confirmed",
    "warnings": ["minor glare on lower edge"],
    "retake_instruction": null
  }
}

```

14. Human Review and Override Rules

The system may allow a supervisor or QA inspector to override an image quality decision, but the override must be logged. Operators should not bypass image quality checks repeatedly without supervisor approval.

Situation	Required Action
Repeated RETAKE_IMAGE result for same product	Send to human inspector after two failed retake attempts.
Possible critical defect visible in poor image	Escalate to human review instead of accepting or rejecting automatically.
Image category mismatch	Ask user to correct category or send to human review.
Model confidence low and image quality warning exists	Do not issue strong final decision; mark human review required.
Supervisor override	Record username, reason, timestamp, and final decision.

15. Data Storage and Audit Requirements

For traceability, the system should store image quality decisions with the inspection result. This helps later debugging when a model prediction is wrong.

- Store original uploaded image path or secure reference.
- Store preprocessed image path if saved.
- Store quality metrics and final quality status.
- Store retake reason or warning message.
- Store model prediction only if inference was allowed.
- Store user override details if quality decision was overridden.
- Do not expose private or sensitive production data in public demos.

16. RAG Retrieval Keywords

The following terms should appear in this document so the RAG retriever can find the correct section when the user asks about poor image inputs or retake requirements.

Topic	Keywords and phrases
General image quality	image quality, image validation, usable image, poor image, quality check, input validation
Blur	blur, blurry image, out of focus, motion blur, camera shake, low sharpness
Lighting	too dark, underexposed, too bright, overexposed, glare, reflection, shadow, poor lighting
Framing	cropped product, product not visible, incomplete image, missing edge, full product visible
Resolution	low resolution, pixelated image, compressed image, minimum resolution
Product category	category mismatch, wrong product category, selected category, bottle image, cable image, screw image

Topic	Keywords and phrases
Operator action	retake image, upload clearer image, human review, pass with warning, escalation
Report fields	image quality status, blur score, brightness status, retake instruction, quality warning

17. RAG Chunking Guidance

For best retrieval quality, chunk this document by section and category. Each category-specific section should be indexed as its own chunk, because users may ask category-specific questions such as "Why was my screw image rejected?" or "What image quality is needed for zipper teeth inspection?"

- Use section titles as metadata: section_name, category, decision_code, issue_type.
- Keep each chunk around one SOP topic, such as blur, lighting, or category-specific retake rules.
- Include product category metadata for bottle, cable, screw, transistor, tile, toothbrush, zipper, and metal_nut.
- Keep table rows together when chunking because the condition and action must remain connected.
- Store document_type = Image_Quality_Check_SOP for all chunks from this PDF.

18. Sample RAG Test Questions

Use these questions to test whether the RAG system retrieves this SOP correctly.

Question	Expected Retrieved Section
Why did the system reject my blurry screw image?	Screw Image Quality Requirements; Blur rules; Frontend messages
What should the operator do if a bottle image has strong glare?	Bottle Image Quality Requirements; Retake rule; Lighting guidance
Can the AI still inspect an image with minor glare?	Image Quality Decision Codes; PASS_WITH_WARNING rules
What report fields should store image quality results?	Image Quality Metrics for Reports; JSON schema
When should a poor image be sent for human review?	Human Review and Override Rules; Agent Rules
What image requirements are needed for zipper teeth inspection?	Zipper Image Quality Requirements
What happens if the uploaded category does not match the product?	Category plausibility; Agent rules; Frontend message
What should be done when OCR text is unreadable?	Agent Rules; Category-specific transistor text guidance

19. Example Scenarios

19.1 Scenario: Blurry screw image

Input: User uploads a screw image where the threads are not visible. Quality result: RETAKE_IMAGE.
System response: "The screw image is too blurry for reliable thread inspection. Please retake the image with the camera focused on the screw threads and include the full screw head, shaft, and tip."

19.2 Scenario: Bottle image with glare

Input: User uploads a bottle image with strong reflection on the sidewall. Quality result: RETAKE_IMAGE or PASS_WITH_WARNING depending on glare size. System response: "Strong glare is hiding bottle surface details. Please adjust the lighting or camera angle and upload again."

19.3 Scenario: Toothbrush bristles cropped

Input: Brush handle is visible, but the bristle head is partially cropped. Quality result: RETAKE_IMAGE. System response: "The brush head and bristles are not fully visible. Please capture the complete toothbrush head because bristle defects cannot be inspected from this image."

19.4 Scenario: Tile image with mild shadow

Input: Full tile is visible, but one corner has mild shadow. Quality result: PASS_WITH_WARNING. System response: "The image is usable, but mild shadow may affect surface defect detection. The report will include an image quality warning."

20. Final Acceptance Checklist

Before the system performs defect detection, the image should pass the following checklist.

Checklist Item	Pass Condition
File readable	Image file opens successfully and is a supported type.
Resolution acceptable	Image has enough pixels for the selected model.
Product visible	Full product or required ROI is visible.
Focus acceptable	Important surface or structural details are sharp.
Lighting acceptable	Image is not too dark, too bright, or strongly reflected.
Background acceptable	Background does not confuse the product boundary.
No major occlusion	No hand, tool, shadow, or object hides the inspection area.
Category correct	Uploaded product matches selected category or is confirmed by user.
Text readable if required	OCR-relevant text is clear enough when label verification is needed.
Quality status recorded	Quality result is stored in report and database.

21. Final Notes

This SOP is designed to improve reliability in a prototype AI inspection assistant. It supports safe decision-making by preventing poor images from entering the defect detection and RAG explanation pipeline. In the final project, the thresholds and acceptance rules should be tuned using real validation images, model confidence results, and feedback from human inspectors.

This document should be used together with the QA Inspection Manual, Defect Severity Guide, Operator Action SOP, Root-Cause Troubleshooting Guide, Inspection Report Template, and Human Review and Escalation Policy.